YTL*AI LABS Z.AI

UM HACKATHON 2026 · FINAL ROUND

RAILMIND

"Malaysia's first AI-native train dispatch system — built to save time, reduce cost, and generate strategic data."

KELANA JAYA LINE | AMPANG LINE | SRI PETALING LINE | POWERED BY Z.AI GLM-5.1

REAL-TIME AI-DRIVEN LRT MANAGEMENT INTELLIGENCE & DISPATCH

THE CHALLENGE

Manual scheduling breaks down at scale

REAL-WORLD SCENARIO

It's Saturday evening. **Blackpink is performing at Stadium Bukit Jalil.** 13,000 fans exit at once. The station master calls OCC. OCC manually estimates the crowd and adjusts the headway. That process takes **10–15 minutes** — platforms reach crush load before a single extra train is deployed.

01

Nounified data view

Weather, events, and emergencies operate in silos. OCC has no single dashboard to assess network-wide load.

BLIND DECISIONS

02

10–15min verbal lag

Station master → phone call → OCC → manual calculation → instruction to drivers. Demand has already shifted by then.

TOO SLOW

03

Relies on memory

Every OCC manager handles events differently. No documented playbook, no institutional learning, no consistency.

HUMAN ERROR

04

3 emergencies, 0 playbook

Signal failure, power failure, overcrowding — each requires a different response, with no automated guidance.

NO PLAYBOOK

OUR SOLUTION

Four steps from problem to updated schedule

1

INPUT

Describe the situation

Staff enter the weather, nearby event, or emergency — via form, free text, image upload, or CCTV snapshot. Any format works.

WEATHER EVENT EMERGENCY CCTV

2

COMPUTE

System calculates 3 options instantly

Amathengine computes Conservative, Moderate, and Aggressive options — load factor, passengers served, and cost for each — in under 2 seconds.

CONSERVATIVE MODERATE AGGRESSIVE

3

RECOMMEND

GLM-5.1 picks one and explains

Z.AI GLM-5.1 reads the situation, weighs all three options, and streams a plain-language recommendation with full reasoning. Staff can always override.

LIVE STREAMING PLAIN LANGUAGE OVERRIDE-READY

4

IMPLEMENT

Apply, save, and hand over

One click applies the option to the full 18-hour schedule. Saved to SQLite — any shift can retrieve it. GLM writes the handover briefing automatically.

ONE-CLICK APPLY SAVED TO DB AUTO BRIEFING

SITUATION INPUT

Four ways to describe any situation

01

Manual Inputs

Simple dropdowns: weather (clear / cloudy / rainy / stormy), event type and crowd size, and emergency type. No typing. No API call. Results are instant.

Weather: Rainy · Event: Concert · Emergency: Overcrowding

02

Free-Text Description

Just type what's happening in plain English or Malay. GLM-5.1 reads the description and automatically extracts weather, event type, crowd size, and any emergency conditions.

"Blackpink concert at Bukit Jalil, ~4,000 extra pax/hr. Rainy weather."

03

Instant Event Recognition

Upload a concert poster or event ticket. OCR + GLM extracts venue, date, start time, and capacity tier automatically — then generates a pre-event adjustment plan.

"Blackpink @ Axiata Arena — 13,000 pax — 19:00–22:00 auto-detected"

04

Real-Time Passenger Detection

Upload a CCTV snapshot of the platform. The computer vision model counts people, estimates crowd density, and feeds that live number directly into the frequency engine.

"CCTV at Bukit Jalil — 340 people detected → 51,000 est. pax/hr"

ESG IMPACT

RM 2.2–3.5 million saved annually through energy optimization

RAILMIND's smart dispatch and energy optimization **avoids RM 2.2–3.5 million per year** in potential carbon tax liability. Every optimized train run is a step toward greener public transit.

Carbon Tax Avoidance

RM 2.2–3.5M/year in avoided liability through smart dispatch energy optimization. Fewer empty trains, less idle burn, lower taxable emissions.

Emission Reduction

Optimized train frequencies reduce idle running and unnecessary energy burn. Data-driven dispatch matches supply to real demand, not guesses.

Sustainable Operations

Data-driven dispatch = fewer empty trains, lower carbon footprint per passenger-km. RAILMIND turns operational efficiency into environmental impact.

ESG SCORE IMPACT: HIGH · CARBON TAX RISK: ELIMINATED · GREEN TRANSIT: ENABLED

REAL-WORLD FEASIBILITY

Zero hardware. Three phases. Fully deployable today.

DEPLOYMENT ROADMAP

1 **Shadow Mode — Kelana Jaya Pilot** Month 1–4

RAILMIND runs parallel to manual OCC decisions with zero operational risk. AI recommendations logged vs actual decisions to prove accuracy before going live.

2 **Full KV Rollout — All 3 LRT Lines** Month 4–12

Expand via existing faregate APIs to Ampang and Sri Petaling. Multi-line coordination at Masjid Jamek, Chan Sow Lin interchanges.

3 **Multimodal & Strategic Layer** Month 12–18

Bus feeder sync via station exit discharge wave prediction. Sovereign Data Package finalized for ECRL and Penang Mutiara LRT negotiations.

WHY IT'S FEASIBLE NOW

Zero Hardware Required

Secure API layer on existing Prasarana faregate & ATS, no physical installation

Staff-First, Human-Approved

Every dispatch decision confirmed by duty managers. RAILMIND recommends, humans approve

Safety Hard-Coded, Not AI-Dependent

Track incidents always = 0 trains. 6 emergency protocols locked in math layer, GLM cannot override

KEY PARTNERS

- Prasarana
- MRT Corp
- MetMalaysia
- MOT / EPU
- Ticket2u
- NACSA

TARGET CUSTOMERS

Three segments, one platform

PRIMARY

Domestic Transit Operators

Prasarana MRT Corp

Transition from manual, reactive dispatching to automated, data-driven frequency management across Kelana Jaya, Ampang, and Sri Petaling lines.

REVENUE MODEL

SaaS licensing fee across the Klang Valley network

SECONDARY

Global Rail Authorities

ASEAN South Asia Africa

State-owned operators in underdeveloped markets use RAILMIND as a "Digital Leapfrog", modernizing manual rail without hardware-heavy debt.

REVENUE MODEL

Export licensing & implementation consultancy

STRATEGIC

Government Negotiators

MOT EPU

The Ministry of Transport and EPU leverage RAILMIND's proprietary "Sovereign Data Packages" to challenge inflated contractor pricing for new projects.

REVENUE MODEL

Data Package licensing for rail project bidding cycles

SENSORY EVOLUTION

Transitioning from Manual Input to Autonomous Ingestion

CCTV Feed

Live visual ground truth

Event Listings

Ticket2u, AXS scrapers

Weather API

MetMalaysia live feed

RAILMIND AI BRAIN

Live CCTV Streaming (YOLOv8)

Removes the 2-minute faregate lag. Real-time crowd detection feeds directly into the Dynamic Frequency Engine as a continuous signal.

Auto-Scraping Event Listings

Ticket2u and event sites scraped 7 days in advance. Auto-generates surge windows and frequency uplift before the crowd arrives.

MetMalaysia API Integration

Triggers frequency caps before the first raindrop hits. Flash flood warnings automatically trigger stormy-weather headway adjustments.

Solving the Last-Mile Gap with Discharge Wave Prediction

THE "WAVE" CONCEPT

Feeder Bus Sync

Rail exit data predicts the surge, station exit discharge wave data forecasts exactly when and how many passengers will leave. Buses are dispatched to meet the wave, not after it has already passed.

Result: Zero last-mile gap. Passengers step off the train and straight onto a waiting bus.

Interchange Coordination

If the Sri Petaling line surges at Bukit Jalil, the Kelana Jaya and Ampang lines adjust headways at Masjid Jamek **proactively,** before the interchange platform floods.

Result: All connected lines respond as one synchronized network, not as isolated silos.

GLOBAL LEAPFROG

Exporting Malaysian Intelligence, Not Just Infrastructure

40+ Nations still on manual rail dispatching , each one a potential RAILMIND deployment

01 Target: 40+ Nations on Manual Dispatch

ASEAN & South Asia: Indonesia, Vietnam, Bangladesh, Pakistan

Urban rail still on manual dispatch with no path to Siemens-level automation.

02 The "Digital Leapfrog"

Zero-hardware architecture modernizes any existing manual rail system. No multi-billion dollar debt of Siemens or Alstom procurement.

03 Revenue Moat: Rail AI Exporter

Transitioning Malaysia from a technology importer to a Rail AI Exporter. A successful Rapid KL deployment becomes the case study that opens every door across 40+ nations.

RAILMIND

DEMO!